Universidad de los Andes | FACULTAD DE INGENIERÍA Y CIENCIAS APLICADAS

Report: Pre Professional Practice 2025

Faculty of Engineering and Applied Sciences,
Universidad de los Andes

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Abstract

The pre-professional internship took place at the Faculty of Engineering and Applied Sciences of the University of the Andes, under the supervision of Professor Sergio Quijada Figueroa. During this period, two complementary lines of work were addressed: the application of artificial intelligence in education and in public administration. In the first stage, a system was designed to optimize the recording of online classes, integrating AI-based transcription and automatic editing tools to improve the efficiency and quality of teaching materials. In the second stage, AuditorIA was developed, a system designed to automate the classification of public purchase orders according to the subheadings of the Ley de Presupuestos, combining heuristic rules and generative language models (Gemini) within a modular ETL architecture.

The system correctly classified 94.2% of the processed purchase orders (n = 1,730), with subheadings 22 Goods and Services (59.2%) and 29 Non-Financial Assets (32.0%) standing out as the main expenditure categories. This result demonstrates the robustness of the implemented hybrid approach and its potential for future automated audits. The experience allowed for the integration of programming, data analysis, and systems design knowledge, strengthening key competencies in critical thinking, planning, and the practical application of engineering in real-world contexts.

1. Introduction

1.1 Company Description

The Faculty of Engineering and Applied Sciences at the University of the Andes is the academic unit dedicated to training engineers with a comprehensive vision, capable of developing and leading projects that contribute to the country's development, addressing the nation's technological and social challenges with a scientific, ethical, and professional foundation. Since 1995, the faculty has progressed to become a university of excellence due to its innovative capacity, integrating teaching with research, which has led to the development of major research projects in collaboration with students.

Image 1.1 Official logo of the Faculty of Engineering and Applied Sciences.

Currently, the Faculty offers undergraduate programs in various engineering fields, such as chemical, industrial, electrical, computer, and civil engineering, as well as graduate programs including master's and doctoral degrees focused on professional specialization. These programs aim to strengthen scientific research and the professional development of its students.

Its team comprises more than sixty members, including academics, researchers, and administrative staff who support academic and outreach activities. The Faculty's organizational structure comprises a Dean in charge of general management, along with Vice-Deans who oversee areas such as academic affairs, finance, postgraduate studies, and student affairs. This structure allows for a hierarchy designed to integrate teaching, research, and outreach.

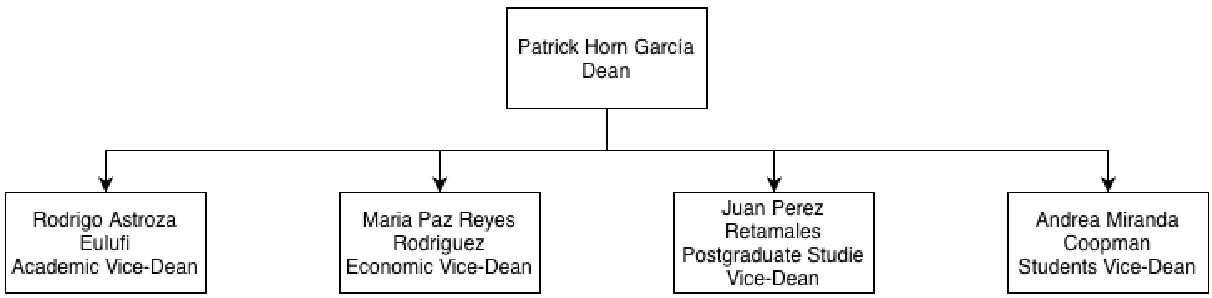

Diagram 1.1. Organizational chart of the authorities of the Faculty of Engineering and Applied Sciences of Universidad de los Andes.

The Faculty maintains its focus on academic excellence, interdisciplinary collaboration, and continuous innovation, fostering active learning environments that enhance the creativity and ongoing commitment of future professionals. In this way, the Faculty fulfills its mission of training well-rounded professionals with ethical values and a commitment to the advancement of science and technology.

1.2 Department/Area Description

The internship took place in the Department of the Master's Program in Management Engineering (MEM), part of the Faculty of Engineering and Applied Sciences at the University of the Andes. This program combines technical and management training, and its director, Sergio Quijada Figueroa, is currently researching new methodologies based on artificial intelligence.

Professor Sergio Quijada, who was my direct supervisor during the Pre-Professional Internship, teaches courses in Simulation, Human Factors Engineering, and Industrial Project Evaluation Design. He is also the Director of the MEM program and a researcher at the Faculty. His growing interest in artificial intelligence has led him to investigate existing technologies and their applications in areas related to public policy and education.

Image 1.2. Supervisor Professor, Sergio Quijada Figueroa

During the internship, I worked directly with Professor Quijada on various exploratory and technological projects. In the first stage, I acted as a technical advisor in designing an effective online class recording plan, proposing AI-based solutions to improve the quality of course materials for the various subjects taught by the professor. This included integrating automated transcription, automatic summary generation, and even an assistant for editing the class recordings. Subsequently, I worked as a researcher on the application of AI-powered public audit automation, investigating the use of Gemini (Google AI) and its various applications, such as its App Canvas environment, for developing a comparative Ley de Presupuestos application. All of this was carried out independently with direct review and supervision from Professor Sergio.

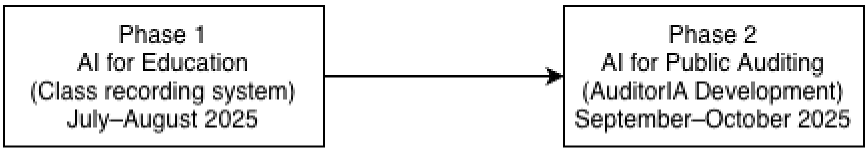

Diagram 1.2 Timeline showing the two main stages of the pre-professional internship.

2. Work Details

2.1 General Activities

During my internship, I developed two lines of work requested by the professor, both focused on the application of artificial intelligence to various tasks of interest to Professor

Sergio, specifically in the areas of education and public policy. My work was academically structured with research guidance and supervision, combining technical, design, and consulting tasks related to technological innovation applied to both education and public policy support.

Initially, the professor asked me to conduct research on the various applications that can be used for recording online classes, with the aim of optimizing his workflow and the quality of his course recordings. This required considering all the technical aspects involved in this audiovisual material and all the professor's needs. To this end, I conducted in-depth research on the different applications available that could improve the professor's workflow. The research began with a preliminary assessment, taking into account the professor's requirements regarding the different online class formats he teaches. In this regard, three types of classes were identified:

1. Classes with screen presentations
2. Classes with a whiteboard
3. Hybrid classes (whiteboard + screen presentation)

For classes with screen presentations, a workflow was developed that primarily uses OBS Studio to record the classes, aiming for better recording quality. For classes with a whiteboard, a workflow was developed depending on the teacher's preferences. The option of a glass whiteboard was considered, which allows for mirroring during editing for proper class development. The use of a digital whiteboard on an iPad/tablet or on graphics tablets (Wacom) connected to a PC was also considered. Finally, for hybrid classes, the use of an iPad/tablet connected to a PC is recommended to allow complete freedom when switching between modes (whiteboard or screen presentation).

For all these options, the AI implementations consisted of tools for use after the class. For transcriptions, the Otter.ai application is recommended, as it has intelligent transcription options. For editing, Clipchamp is recommended, as it has an AI application for intelligent class editing. However, it should be noted that these applications require payment for full use.

Whiteboard Class	**Screen Presentation**	**Hybrid (Screen + Whiteboard)**
Use Glass board setup (mirrored editing) or a tablet/graphic tablet (Wacom, iPad) connected to PC.	Use OBS Studio to capture screen and audio. Focus on slide presentation and system demonstrations.	Combine tablet mirroring with PC screen capture for flexible teaching mode switching.

Table 2.1 Recording configurations and workflows for different class types.

Subsequently, the practice shifted towards the field of artificial intelligence applied to the analysis of public data, specifically towards the proper use of public resources. In this context, the professor challenged me to investigate how artificial intelligence could be used to automate the auditing of public sector budget execution. Initially, the professor asked me to investigate the potential of Gemini AI and its Google AI Studio environment, which includes App Canvas, a generative AI tool capable of creating prototypes of web tools. A first prototype was created comparing Ley de Presupuestos indices from different years. However, the platform's technical limitations motivated the development of an independent solution in Python.

Based on this need, I began building my own analysis system, designing the first functional script prototypes that launched the project. The first, auditor_IA.py, connects to the ChileCompra API to download purchase order data from the Servicio de Salud Talcahuano. The second, analisis_auditorIA.py, processes the obtained CSV files, classifies them into their respective subheadings using a Gemini prompt, and generates the first comparisons between the subheadings of the Ley de Presupuestos and the corresponding purchase orders, thus generating the first analysis metrics.

However, this also had its limitations, as the metrics obtained from classification using only artificial intelligence were absurd, leading to the conclusion that classifications using only Gemini's AI vary significantly depending on the description of the purchase orders. In addition to the monetary cost limitation of classifying all purchase orders with this artificial intelligence.

As a result, the need arose for further research into how to obtain more accurate classifications. This research led to a new model with a more complete and robust architecture that incorporates heuristic rules and Language Modeling Languages (LML) such as Gemini for the automatic classification of orders according to the subheadings of the Public Sector Ley de Presupuestos. Therefore, a pipeline was developed with the following structure:

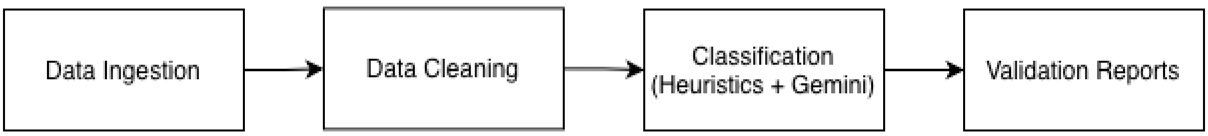

Diagram 2.1. General architecture of the *AuditorIA* system pipeline.

This repository allows for the unification, cleaning, and classification of data from all months of a year. However, the classification is performed using heuristic rules and, ultimately, Gemini for purchase order descriptions that could not be classified using the heuristics, in order to reduce costs and achieve a higher accuracy rate.

Throughout this process, interaction with the professor was constant. We reviewed progress, defined improvement strategies regarding limitations, and analyzed potential scope. All of the above involved applying prior knowledge of programming, systems design and logical reasoning, as well as communication skills to explain the technical results to the professor.

2.2 Most Challenging Activity

During my pre-professional internship, developing the budget audit system was the most complex and challenging activity, both technically and methodologically. The project initially aimed to compare two consecutive Ley de Presupuestos but evolved radically into automating the classification and analysis of public procurement orders according to the subheadings of the Public Sector Ley de Presupuestos, applying a hybrid approach that combined heuristic rules with generative artificial intelligence models.

What began as a simple experimental idea, with two Python scripts (auditorIA.py and analisis_auditorIA.py), grew into a robust modular architecture with controlled data flows and continuous validation. The process involved confronting real limitations in design, scalability, structure, and quality control.

In developing the two initial scripts, auditor_ia.py, responsible for connecting to the ChileCompra API and downloading data from the Talcahuano Health Service, and analisis_auditoria.py, responsible for processing the files, performing the initial classifications using Gemini, and calculating basic metrics for comparison with the Ley de Presupuestos, there were no major difficulties. However, these programs were linear, and the artificial intelligence's classifications of purchase orders varied considerably. Therefore, one of the first challenges arose: finding a way to achieve a high degree of accuracy in classifying according to the Ley de Presupuestos's subheadings.

Because of the above, I dedicated considerable time to researching how data science projects structure their pipelines. I analyzed different methodologies to find the best way to address this problem and ensure the project's success. To this end, I discovered the ETL (Extract, Transform, Load) pipeline development methodology, which allows a project to be segmented into independent but interconnected modules. Based on this research, I began developing a complete restructuring of the previously very simple system, focused on improving readability, reproducibility, scalability, and quality control of the results. This led to a modular architecture that divided tasks by purpose:

```
presu-auditor/
├── data/
│   ├── raw/       → CSV originales desde Mercado Público
│   ├── curated/   → Datasets limpios y predicciones finales
│   └── staged/
│       └── review/ → Reportes de control de calidad
├── src/
│   ├── ingesta/      → Scripts de carga y limpieza
│   ├── clasificador/ → Scripts de clasificación heurística e híbrida
│   └── pruebas/      → Scripts de QA y validación
├── .env              → Clave API y configuración de Gemini
└── requirements.txt   → Dependencias del entorno Python
```

Image 2.1 Modular structure of the *AuditorIA* project based on the ETL methodology.

This restructuring allowed the system to be divided into three main stages:

1. Data ingestion and cleaning, to extract, unify, and clean the CSV files containing purchase orders from the API, in order to optimize their use in subsequent stages.
2. Classification, which is initially based on regular expressions, and if the regular expressions do not match the purchase order descriptions, it relies on the Gemini language model.
3. Testing and quality control, to validate the accuracy and consistency of the predictions, generating automatic reports with performance metrics.

In addition, two environment configuration files were added to this structure: the .env file, which manages the keys and configurations for the ChileCompra and Gemini APIs, and the requirements.txt file, which configures the portability of the Python environment. This new architecture resolves data disorganization issues, which, with the previous structure, led to the loss of purchase orders and, consequently, potential biases in the audit. Furthermore, it transformed the project into a modular, scalable, and auditable system, applying engineering principles within a context of public data analysis.

However, implementing the new pipeline brought with it a series of technical challenges. First, optimizing access to the ChileCompra API, which had response time limitations and restrictions on search parameters. To resolve this, it was necessary to create error handling and automatic retry functions, as well as a caching function to avoid overloading requests that could result in API errors.

Another important aspect I had to manage was classification, since initially, because it was only performed by Gemini, many purchase orders were incorrectly classified. Additionally, using the Gemini API involves a significant financial cost. It is for this reason that I had to design the hybrid classifier, where I developed a set of heuristic patterns abstracted from the official DIPRES budget classification. These patterns identified keywords associated with categories such as "Personnel," "Goods and Services," and others. However, in many cases, the descriptions did not match the patterns, so an additional layer of generative artificial intelligence was implemented. This layer intervened when the heuristics failed to classify with sufficient confidence.

This classifier works based on confidence; that is, if the heuristic match has low confidence, the text is sent to the LLM model to suggest a likely category. The final confidence level is also recorded, as well as whether the prediction source was the heuristics or the LLM model, in order to validate and track the reason for the classification in the respective subheading.

On the other hand, I made some mistakes during the development of my project. First, I didn't implement a version control system from the beginning, which sometimes resulted in the loss of progress in the initial stages of the project. Therefore, I started using Git and GitHub to establish documented commits that would allow me to track and visualize this progress. I also made the mistake of giving new CSV files with results similar names, which led to confusion and the deletion of some useful files. Another crucial error in the development of this system was not constantly and thoroughly reviewing the generated CSV files. While I checked that they were generally fine, some files remained uncleaned, which caused the classifier to become confused and unable to correctly link purchase orders to their respective subheadings. Finally, the most significant initial error, which I mentioned earlier, was overconfidence in the LLM. As I described, Gemini was initially solely responsible for classifying purchase orders, leading to unexpected results. However, all these errors were addressed promptly in the scripts, ensuring these files continued through the pipeline correctly.

Despite all the limitations and challenges, the final result was a functional, robust, and structured system capable of automatically processing and classifying public purchase orders, thus reducing and assisting with manual analysis in budget audits. Furthermore, the modular design allows for the future incorporation of supervised models to develop larger-scale budget audit projects.

3. Analysis

3.1 Company Analysis

The Faculty of Engineering and Applied Sciences at the University of the Andes offers an academic environment focused on applied innovation, interdisciplinary research, and professional training with a technological focus. While the Faculty's structure includes various areas and programs, my internship took place within a more specific context: directly with Professor Sergio Quijada Figueroa, who leads research projects in diverse fields, such as simulation and, in this case, artificial intelligence applied to education.

This academic context allowed for personalized work with the professor, where productivity is primarily measured by the generation and exchange of knowledge with the professors. They provide students with more professional perspectives on the design and creation of technology-based solutions, fostering innovation and critical thinking among students. This internship was conducted both in the computer lab and on my personal computer, resulting in individual research with the professor's collaboration. He fulfilled his role as a methodological guide, contributing his knowledge of engineering tools, directing the research, and reviewing my results.

His ability was evident in his flexibility and openness to new lines of research. The professor gave me the opportunity to freely propose a design that would solve the problem at hand, which in this case resulted in Auditing. This demonstrates the trust that the faculty members place in their students, fostering their work and capacity for innovation, focusing their research on their interests but always with the support and guidance of the professors. This educational system promotes innovation, self-directed learning, and the application of knowledge acquired in the various engineering courses offered at the University of the Andes, and it also motivates students toward research.

From a human resources perspective, Professor Sergio Quijada guided, directed, and supervised this research, drawing on his extensive experience in various fields of engineering. Their guidance and expertise allowed me to apply knowledge from various university courses to this project, making their support fundamental to its development. They provided me with insights into how projects of this nature are carried out, offering continuous feedback and assisting me in making decisions regarding the difficulties encountered during the project.

Academically, the Faculty is characterized by its diverse group of researchers who promote high-impact projects. These projects are geared towards solving real-world problems by utilizing the knowledge provided by the university and emerging technologies. In this sense, the project I undertook falls within a line of work that seeks to modernize public administration through artificial intelligence tools, which have been progressively integrated into the curriculum.

In short, the environment in which my pre-professional internship took place is characterized more by its multidisciplinary capacity for research and adaptation than by its infrastructure. Because the practical experience was conducted directly with the professor, it fostered direct communication, enabling quick decision-making and continuous motivation and learning. This demonstrates that the Faculty is an environment that aims to train engineers capable of innovating, researching, and proposing solutions to real-world problems, integrating all the knowledge acquired throughout the process.

3.2 Work Analysis

During my pre-professional internship, I developed the AuditorIA system, a tool designed to automate the classification of public purchase orders according to the subheadings of the Ley de Presupuestos. This project falls within the new line of research led by Professor Sergio Quijada Figueroa and seeks to demonstrate how artificial intelligence can contribute to public policy. This internship provided me with an opportunity to apply various skills I have acquired throughout my studies, integrating programming, data analysis, and even systems design.

Developing AuditorIA opens a new methodology for budget analysis based on access to public data and automation applied with artificial intelligence. Prior to this, no systems existed that incorporated the downloading, cleaning, and automated classification of purchase orders from the ChileCompra portal. This new system is enabling a more efficient, reproducible, and scalable data processing workflow. In this way, the project contributes to Professor Sergio Quijada's research on the integration of artificial intelligence in public policy.

The work also involved a technical and methodological redesign process. In its initial version, the system consisted of two simple Python scripts that performed specific functions. However, the limitations of this structure were evident, as the workflow did not allow for error control or the separation of stages. To solve this problem, I had to research various methodologies and found that the ETL (Extract, Transform, Load) approach was the most relevant for this project. This allowed for the development of a modular system, which improves code organization, efficiency, and facilitates the traceability of results.

Among the decisions that enabled the project's development, one of the most relevant was the creation of the hybrid classifier, which combines heuristic patterns, through regular expressions, with the Gemini AI language model. This allowed me to realize that by relying entirely on the artificial intelligence model, the model generates inconsistent results and high operating costs. By integrating heuristics, it was possible to achieve a more reliable classification and reduce the workload on artificial intelligence, improving the system's accuracy and lowering the error rate in initial tests, all at a lower cost.

Quantitatively, the system processed a total of 1,729 purchase orders, of which 94.2% were correctly classified according to the Ley de Presupuestos subheadings, while the remaining 5.8% were unclassified.
The majority of the orders (59.2%) were assigned to subheading 22_Goods_and_Services, and 32.0% to subheading 29_Non_Financial_Assets, which aligns with the typical expenditure structure of a healthcare service.

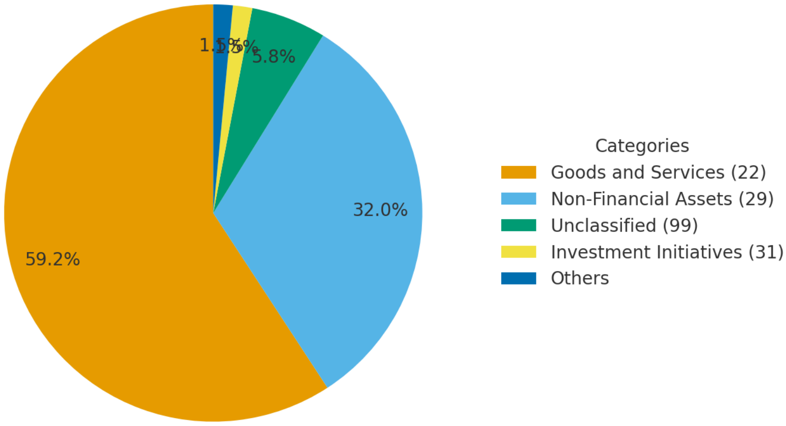

Figure 3.1 Distribution of classified purchase orders by budget subheading

Furthermore, my critical analysis during the project's development allowed me to identify various errors that I was able to transform into valuable learning experiences. First, the lack of a version control system caused me to lose significant progress, so I realized that implementing systems like Git and GitHub is crucial when creating pipelines of this type, to maintain an organized and documented record of progress. Furthermore, the initial standardization of CSV files made me realize the importance of order when processing data, as without it, much important information can be lost. Finally, overconfidence in artificial intelligence models led to misclassifications, so it's important to be critical and understand what you're doing before applying these models to your projects.

From a professional perspective, the internship allowed me to develop advanced technical and analytical skills, as well as engineering competencies such as planning, validation, error control, and the documentation and creation of workflows. I understood that projects must be approached holistically, considering that the project must be sustainable and scalable over time.

In summary, the AuditorIA course allowed me to directly experience the entire design process of a technological system, from its inception to its implementation. All the decisions I made in conjunction with the professor contributed to strengthening my critical thinking and engineering judgment. This practical experience became a formative one that solidified my understanding of how an engineer's development requires a broad perspective on the solutions they develop, extending beyond the purely technical aspects.

4. Personal Reflection

My pre-professional internship was a unique opportunity for me, allowing me to realize the comprehensive nature of an engineer's role. Working with Professor Sergio Quijada Figueroa enabled me to delve into and research two areas that greatly interest me: public policy and artificial intelligence. This research environment gave me a different perspective on what it means to be an engineer. I realized that every advancement depends directly on my technical judgment, analytical skills, and discipline in achieving the proposed objectives.

From a professional perspective, the internship allowed me to apply knowledge acquired during my studies, such as programming and data analysis, in a real-world context that can have a significant impact on society, given the considerable influence of public policy. Furthermore, I understood that solutions to problems must be developed in a structured way, considering aspects like scalability so that systems can endure over time and continue to be developed and improved.

As a researcher, this experience taught me the importance of perseverance, organization, and critical thinking in the work we do. There were times when I seemed to be making no progress, and I had to better manage my time to keep up with the progress the professor was asking for. Managing frustration is also crucial when things don't go as planned. I realized that mistakes don't always mean setbacks; they are opportunities to improve my approach and find solutions and alternatives. Furthermore, communication is essential in projects like this. It allowed me to work effectively with the professor, and we understood each other well regarding his research needs and the progress I was proposing.

In conclusion, this internship was a turning point in my engineering development. It allowed me to explore an area of engineering I was unfamiliar with and which I initially feared due to its inherent challenges. However, this experience allowed me to strengthen my knowledge and explore new skills I hadn't previously used.